

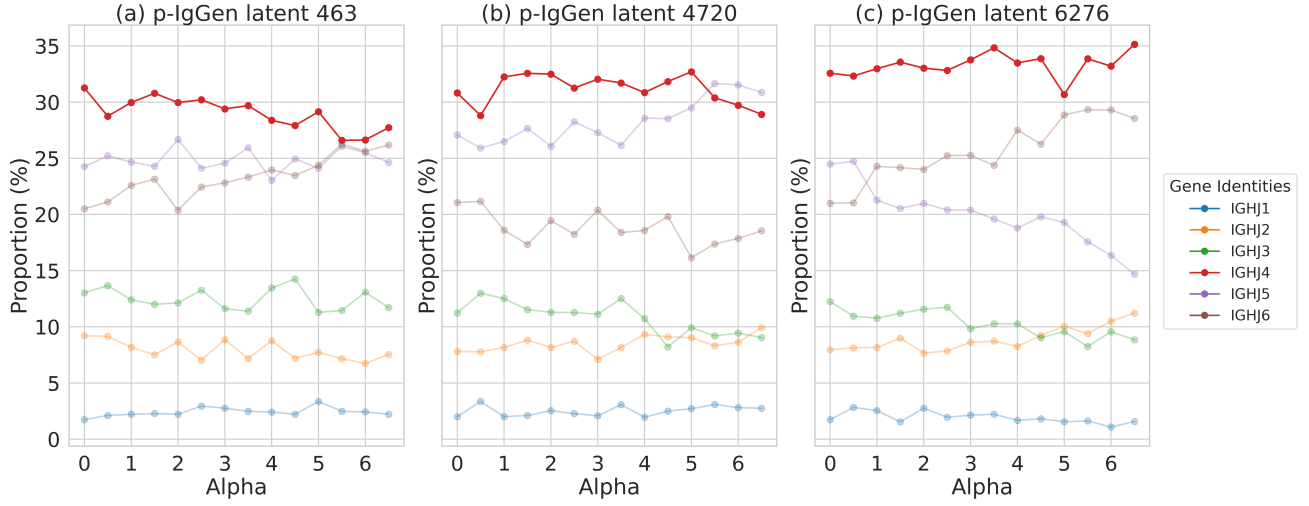


Figure 5. Results of IGHJ4 feature steering for TopK latent 463 (a), 4720 (b), 6276 (c). Y-axis shows the proportion of generated sequences. Plots are coloured by heavy J gene identity. X-axis shows the steering factor used (alpha). Results are for a library of 1000 p-IgGen-generated sequences. For each latent tested (a-c), steering did not result in a predictable change in library composition.

Table 4. Layer 3 feature counts and maximum F_1 -scores for IGHJ genes. A latent is counted as a feature if it satisfies $F_1 > F_{1,\text{all}+} + 0.2$ for any tested activation threshold, where $F_{1,\text{all}+} = \frac{2\pi}{1+\pi}$ and π is the validation prevalence of the positive class.

Gene	Number of features	Max F_1	$F_{1,\text{all}+} + 0.2$
IGHJ1	1	0.366	0.244
IGHJ2	6	0.758	0.393
IGHJ3	17	0.866	0.364
IGHJ4	1	0.752	0.713
IGHJ5	0	0.486	0.598
IGHJ6	16	0.866	0.517

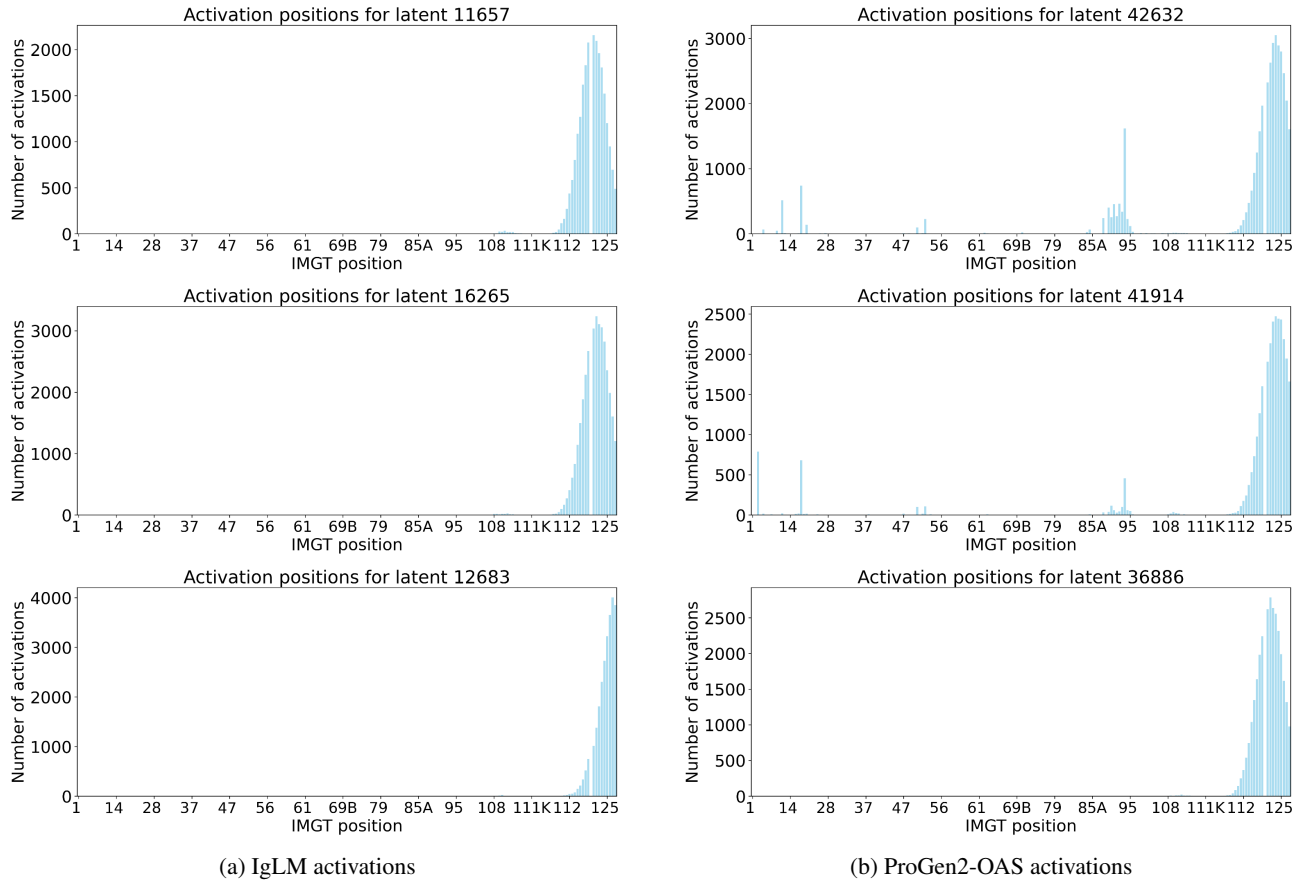
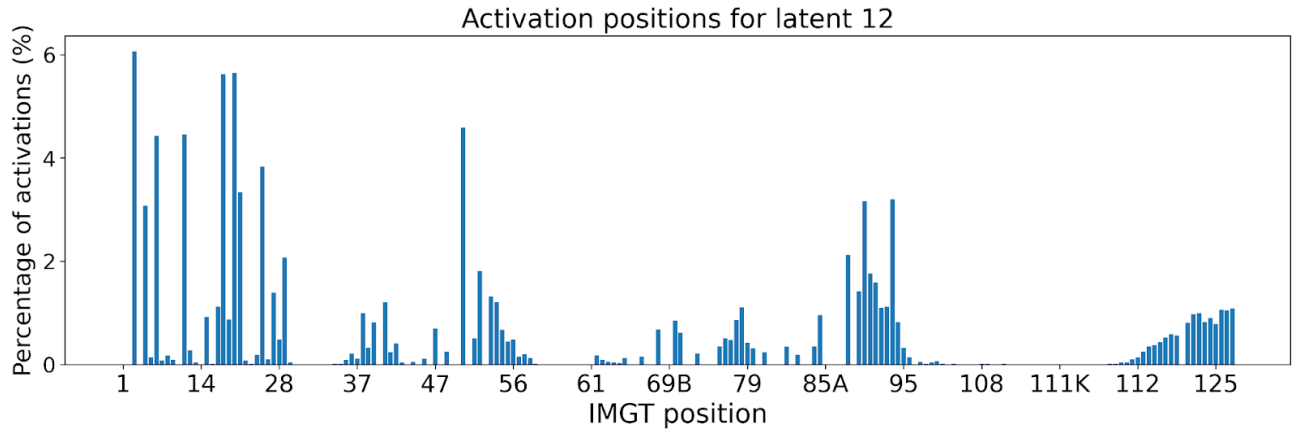
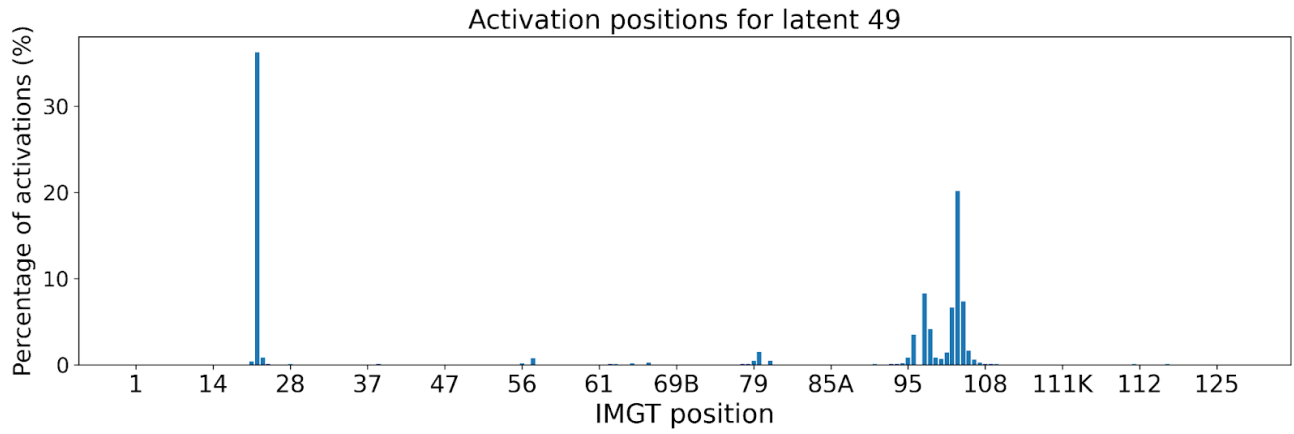


Figure 6. IMGT activations of top three TopK IgLM (a) and ProGen2-OAS (b) IGHJ4 features. The IMGT positions are shown on the x-axis. Percentage of total activations on any given position across validation IGHJ4 sequences is shown on the y-axis. Latent activations show a distribution near the end of the heavy chain when aligned based on IMGT numbering.



(a) Activation pattern of top positively correlated O-SAE latent for p-IgGen



(b) Activation pattern of top negatively correlated O-SAE latent for p-IgGen

Figure 7. IMGT activations of p-IgGen O-SAE latent 12 (positively correlated) (a) and 49 (negatively correlated) (b). Activation patterns of both latents show scattered distribution across the range of IMGT positions.